



Employee Listening & Organizational Health Analytics

Executive Summary

This analysis examines employee experience across six organizational constructs—Engagement, Manager Effectiveness, Psychological Safety, Growth & Development, Workload Sustainability, and Belonging—and evaluates their relationship with employees' intent to stay. The analysis uses 4,344 usable survey responses, with 4,071 valid responses for the intent-to-stay outcome.

Overall employee-experience scores were relatively consistent and centered around the midpoint of the five-point scale. Growth & Development had the highest overall mean score (3.02), while Belonging had the lowest (2.98). Differences across business units were generally modest, although department-level analysis identified larger variation in selected areas, particularly Growth & Development and Manager Effectiveness.

Intent to Stay had an overall mean of 3.02 (SD = 1.22). All six employee-experience constructs were positively associated with intent to stay. Engagement showed the strongest association ($r = .529$), followed by Manager Effectiveness ($r = .524$) and Belonging ($r = .503$).

A multiple linear regression model using the six employee-experience constructs explained approximately 58% of the variance in Intent to Stay on the held-out test data ($R^2 = .582$; RMSE = .794; MAE = .640). Engagement had the largest standardized association with intent to stay, while all six constructs remained positive and statistically significant in the final model. Adding job level, manager status, work arrangement, and tenure did not significantly improve the model fit ($p = .700$).

Taken together, the findings suggest that employee experience—particularly Engagement, Manager Effectiveness, and Belonging—is more informative for understanding intent to stay in this dataset than basic employee characteristics alone. These findings can help organizational leaders prioritize areas for further investigation and targeted employee-experience interventions.

Business Question & Objectives



What aspects of the employee-experience at Meridian are most strongly associated with intent to stay, and which organizational groups may need additional support?

–Business Question



Objectives

This analysis was designed to accomplish the following:

1. Assess the overall employee-experience across six core constructs:
 - Engagement
 - Manager Effectiveness
 - Psychological Safety
 - Growth & Development
 - Workload Sustainability
 - Belonging
2. Identify organizational differences across business units and departments to determine where employee-experience scores vary most.
3. Examine the relationship between employee-experience and intent to stay by assessing the strength of associations between each construct and the outcome.
4. Develop and evaluate a predictive model of intent to stay using the six employee-experience constructs.
5. Determine whether basic employee characteristics—job level, manager status, work arrangement, and tenure—add meaningful predictive information beyond the employee-experience measures.

Survey & Data Overview

Organizational Context

Meridian Workforce Solutions is a fictional workforce and people-analytics consulting organization used as the business context for this portfolio project. The analysis simulates an internal employee-listening engagement in which Meridian evaluates its own workforce experience and identifies factors associated with employees' intent to stay.

Meridian employs approximately 3,600 employees across three business units: Client Operations, Corporate Services, and Technology & Product. The organization includes functions such as Customer Support, Implementation, Service Delivery, Finance, Human Resources, Sales & Marketing, Data & Analytics, Engineering, and Product.

All employee and survey data used in this analysis are synthetic and were created for demonstration purposes. The project is designed to simulate the type of employee-listening and organizational-health analytics engagement that an internal people analytics team might deliver to organizational leadership.

Survey Design

The employee-listening survey was designed to assess six dimensions of the employee experience: Engagement, Manager Effectiveness, Psychological Safety, Growth & Development, Workload Sustainability, and Belonging. Each construct was measured using multiple survey items and summarized into a construct-level score.

The survey also included an intent-to-stay measure, which serves as the primary outcome for the predictive analysis. Organizational and employee characteristics, including business unit, department, job level, manager status, work arrangement, and tenure, were also available for organizational comparisons and model evaluation.

The analysis uses a five-point scoring framework, with higher scores representing more favorable responses.

Analytical Population

The processed survey dataset contains 4,344 usable responses. Of these, 4,071 respondents provided valid intent-to-stay scores and were available for analyses involving that outcome.

Data Quality & Survey Validation

Response usability and missingness

The processed dataset contains 4,344 usable survey responses. Survey-response records were checked for duplicate employee-wave combinations, expected survey items, and valid five-point Likert response values. All survey items were constrained to the intended 1-5 response scale. Intent-to-stay analyses were based on 4,071 valid observations because 273 usable survey records had missing intent-to-stay values.

Scale reliability

The six multi-item employee-experience constructs demonstrated acceptable internal consistency, with Cronbach's alpha values ranging from 0.754 to 0.780. Item-level diagnostics did not indicate that removing individual items would improve reliability. These results support combining the items within each construct into composite scale scores.

Construct validity

Factorability diagnostics supported further examination of the employee-experience measurement structure. The overall KMO was 0.90, while Bartlett's test of sphericity was highly significant ($\chi^2 = 29,381.6$, $df = 276$, $p < .001$). Parallel analysis recommended six factors, consistent with the six

constructs specified during survey development. Exploratory factor analysis subsequently showed a clean six-factor structure with strong within-construct loadings and minimal cross-loadings.

Reporting thresholds and privacy

To reduce the risk of identifying individual employees, survey results were subject to a minimum reporting threshold of 10 usable responses for organizational groups. Groups with fewer than 10 usable responses would be suppressed rather than reported. Groups meeting the threshold were considered reportable, with additional caution applied where appropriate.

In this analysis, all business units and departments exceeded the minimum threshold, so no organizational groups required suppression. The threshold is intended as a basic reporting safeguard; in a real employee-listening program, additional protections would be needed when examining small demographic or organizational intersections that could increase re-identification risk.

Employee Experience Findings

Overall Employee-Experience

Overall employee-experience scores were tightly clustered around the midpoint of the five-point scale. Growth & Development had the highest mean score (3.02), followed by Manager Effectiveness and Psychological Safety (3.01 each). Belonging had the lowest mean (2.98), while Engagement and Workload Sustainability were both approximately 3.00.

The relatively narrow range of overall scores suggests that no single employee-experience dimension stands dramatically apart from the others at the organization-wide level. Instead, the results indicate a broadly moderate employee experience with relatively similar levels across the six measured dimensions.

Table 1. Overall employee-experience construct scores

Construct	Mean	SD
Growth & Development	3.02	1.10
Manager Effectiveness	3.01	1.09
Psychological Safety	3.01	1.11
Workload Sustainability	3.00	1.09
Engagement	3.00	1.11
Belonging	2.98	1.09

Note. Scores use a five-point scale, with higher values representing more favorable responses. SD = standard deviation.

Business-Unit Differences

Differences across business units were generally modest. Corporate Services had the highest scores for Growth & Development (3.07), Psychological Safety (3.06), Workload Sustainability (3.03), and Engagement (3.01), while Technology & Product had the highest Manager Effectiveness (3.04) and Belonging scores (3.01). The small magnitude of these differences suggests that employee-experience patterns are broadly consistent across the three business units.

Table 2. Employee-experience scores by business unit

	Client Operations	Corporate Services	Technology & Product
Usable Responses	1848	849	1647
Engagement	2.99	3.01	3.00
Manager Effectiveness	3.00	2.99	3.04
Psychological Safety	3.00	3.06	3.00
Growth & Development	3.00	3.07	3.01
Workload Sustainability	3.00	3.03	3.00
Belonging	2.96	2.96	3.01

Note. Scores use a five-point scale, with higher values representing more favorable responses.

Department-Level Differences

Department-level analysis revealed greater variation than business-unit comparisons. Growth & Development showed the largest spread across departments, with Finance scoring highest (3.16) and Service Delivery lowest (2.94). Manager Effectiveness showed the second-largest spread, from 3.12 in Finance to 2.94 in Sales & Marketing. These department-level differences provide more specific areas for organizational follow-up than the relatively small business-unit differences.

Table 3. Department-level variation in employee-experience scores

Construct	Range	Highest Score	Lowest Score	Highest Department	Lowest Department
Growth & Development	0.22	3.16	2.94	Finance	Service Delivery
Manager Effectiveness	0.18	3.12	2.94	Finance	Sales & Marketing
Belonging	0.12	3.02	2.90	Product	Finance

Construct	Range	Highest Score	Lowest Score	Highest Department	Lowest Department
Psychological Safety	0.11	3.09	2.98	Human Resources	Data and Analytics
Workload Sustainability	0.10	3.05	2.95	Product	Service Delivery
Engagement	0.06	3.03	2.97	Sales & Marketing	Service Delivery

Range = highest department score - lowest department score. Note. Scores use a five-point scale, with higher values representing more favorable responses.

Intent to Stay

Overall Intent to Stay

Across the 4,071 respondents with valid intent-to-stay scores, the mean intent-to-stay score was 3.02 (SD = 1.22) on the five-point scale. This places overall intent to stay near the midpoint of the response scale, indicating a mixed rather than uniformly high level of stated intention to remain with the organization.

Because intent to stay is the primary outcome of the analysis, the next step was to examine whether the six employee-experience dimensions were associated with employees' stated intention to remain.

Employee-Experience Associations with Intent to Stay

Table 4. Correlations between employee-experience constructs and intent to stay

Construct	Correlation
Engagement	0.529
Manager Effectiveness	0.524
Belonging	0.503
Psychological Safety	0.498
Growth & Development	0.448
Workload Sustainability	0.426

All six employee-experience constructs showed positive associations with intent to stay. Engagement had the strongest observed association ($r = .529$), followed by Manager Effectiveness ($r = .524$) and

Belonging ($r = .503$). Workload Sustainability showed the weakest of the six associations, although its relationship with intent to stay remained positive ($r = .426$).

These correlations indicate that employees reporting more favorable experiences also tended to report stronger intentions to stay. However, correlations alone do not establish that improving a particular employee-experience dimension would cause employees to remain with the organization.

Predictive Modeling

Modeling Approach

A multiple linear regression model was developed to evaluate the combined relationship between the six employee-experience constructs and intent to stay. The six constructs were entered simultaneously as predictors, allowing the model to estimate each construct's association with intent to stay while accounting for the other five constructs.

The modeling dataset contained 4,071 observations with complete data for the outcome and six predictors. The data were divided into an 80% training set ($n = 3,256$) and a 20% testing set ($n = 815$). Model performance was evaluated on the held-out testing data to assess how well the model generalized beyond the observations used to estimate its coefficients.

Baseline Model Performance

The baseline six-construct model explained approximately 58% of the variance in intent to stay. Performance was highly consistent between the training and testing datasets, with an R^2 of .577 in training and .582 in testing. Testing RMSE was .794, while testing MAE was .640.

Table 5. Baseline model performance

Dataset	R^2	RMSE	MAE
Training	0.577	0.791	0.634
Testing	0.582	0.794	0.640

Note. RMSE = root mean squared error; MAE = mean absolute error.

Multicollinearity

Variance inflation factor (VIF) diagnostics ranged from 1.18 to 1.34, indicating very low multicollinearity among the six employee-experience predictors. The predictors therefore provided sufficiently distinct information for their individual regression coefficients to be interpreted without substantial inflation from predictor overlap.

Employee Characteristics

A second model added job level, manager status, work arrangement, and tenure to determine whether basic employee characteristics provided additional explanatory value beyond the six employee-experience constructs. The expanded model did not significantly improve fit compared with the baseline model, $F(8, 3241) = 0.69$, $p = .700$.

This result suggests that, within this dataset, the employee-experience measures captured substantially more of the model's predictive information than these basic employee characteristics.

Final Model

Because adding job level, manager status, work arrangement, and tenure did not significantly improve model fit, the six-construct baseline model was retained as the final predictive model.

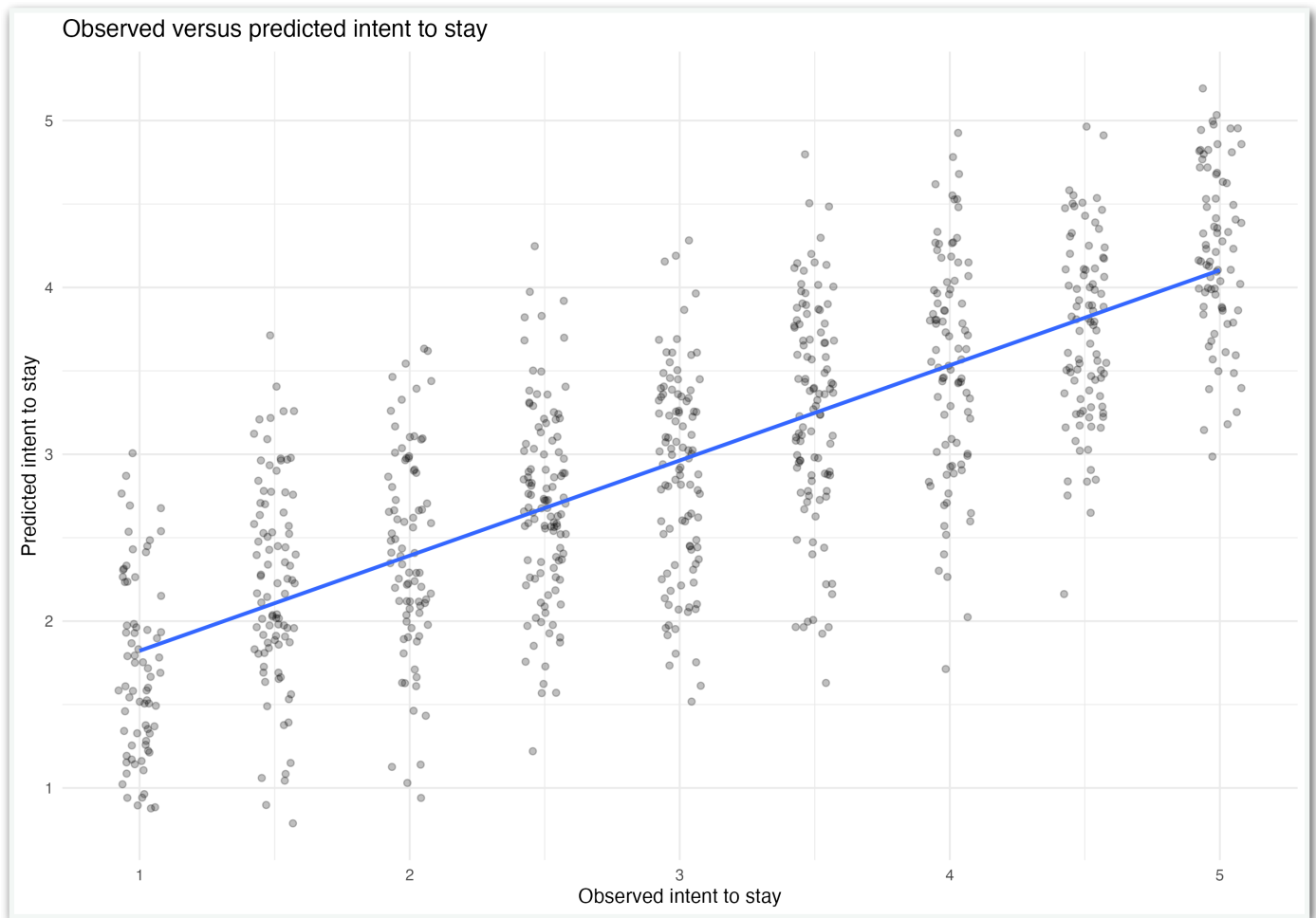
In the final model, all six employee-experience constructs had positive and statistically significant coefficients. Engagement had the largest coefficient ($b = .272$, 95% CI [.245, .300]), followed by Manager Effectiveness ($b = .238$, 95% CI [.209, .266]) and Belonging ($b = .227$, 95% CI [.198, .255]).

Psychological Safety, Growth & Development, and Workload Sustainability also remained positive and statistically significant.

Table 6. Final intent-to-stay regression coefficients

Construct	Estimate (b)	95% CI	p-value
Engagement	0.272	[0.245, 0.300]	<0.001
Manager Effectiveness	0.238	[0.209, 0.266]	<0.001
Belonging	0.227	[0.198, 0.255]	<0.001
Psychological Safety	0.197	[0.169, 0.226]	<0.001
Growth & Development	0.189	[0.162, 0.216]	<0.001
Workload Sustainability	0.171	[0.144, 0.198]	<0.001

Unstandardized regression coefficients (b) from the final model predicting intent to stay. Confidence intervals are 95%; p-values indicate statistical significance. Positive coefficients indicate that higher employee-experience scores were associated with higher intent-to-stay scores.

Figure 1. Observed versus predicted intent to stay on the held-out test set

Predicted values are based on the final regression model evaluated on the held-out test set (n = 815). The figure compares observed intent-to-stay scores with model predictions.

Organizational Implications

Based on the findings, the analysis points to several implications for how Meridian could prioritize employee-listening follow-up.

First, employee-experience appears more informative than basic employee characteristics for understanding intent to stay. The six employee-experience constructs were positively associated with intent to stay, and adding job level, manager status, work arrangement, and tenure did not significantly improve the model. This suggests that organizational leaders may gain more actionable insight by examining the employee-experience itself rather than relying primarily on broad employee characteristics.

Second, Engagement, Manager Effectiveness, and Belonging emerged as the strongest areas of association with intent to stay. Engagement had the strongest standardized association, followed by

Manager Effectiveness and Belonging. These areas may therefore warrant particular attention when considering interventions intended to strengthen the employee-experience.

Third, organizational-level averages can mask more localized variation. Business-unit differences were relatively modest, whereas department-level analysis showed greater variation in selected constructs, particularly Growth & Development and Manager Effectiveness. This suggests that follow-up efforts may be more useful when targeted toward specific organizational groups rather than applying the same intervention uniformly across the organization.

Finally, the findings support using employee listening as an input to further investigation rather than as a standalone decision tool. Lower-scoring departments or constructs should be treated as signals for additional inquiry, not automatically as evidence of an organizational problem. **In a real employee-listening program**, leaders could combine survey findings with qualitative feedback, operational context, and subsequent survey waves before determining appropriate action.

Recommendations

Collectively, the findings suggest that the following recommendations would be appropriate for a simulated employee-listening program at Meridian Workforce Solutions.

Prioritize Manager Effectiveness and Engagement

Because Engagement and Manager Effectiveness showed the strongest associations with intent to stay, Meridian could prioritize interventions that strengthen day-to-day management practices and employee engagement. Potential areas for follow-up include manager communication, feedback, support, clarity of expectations, and opportunities for employees to contribute.

Strengthen Belonging and Connection

Belonging was also strongly associated with intent to stay. Meridian could examine whether employees feel respected, included, able to be authentic at work, and connected to their teams. Follow-up listening could help identify which aspects of Belonging are most relevant to different employee groups.

Target Development Opportunities where Department-Level Gaps are Larger

Department-level analysis showed greater variation in Growth & Development and Manager Effectiveness than was observed across business units. Meridian could therefore use department-level results to identify areas for further investigation and consider targeted development resources, manager support, or career-development initiatives rather than applying organization-wide interventions uniformly.

Use Targeted Follow-up Listening Before Taking Action

Lower-scoring results should be treated as signals for further investigation rather than definitive evidence of an organizational problem. Meridian could follow survey findings with qualitative feedback, manager discussions, focus groups, or subsequent survey waves to better understand the underlying employee-experience before implementing interventions.

Maintain Privacy-Conscious Reporting

Any future employee-listening analysis should continue to apply minimum reporting thresholds and avoid exposing results for very small groups. Additional safeguards should be considered when multiple organizational or demographic characteristics are combined, particularly where small groups could increase the risk of identifying individual respondents.

Important: Because this project uses synthetic data, these recommendations demonstrate how an analyst could translate employee-listening findings into organizational actions; they are not recommendations for a real workforce.

Limitations

There are several limitations surrounding this analysis that should be considered when interpreting the findings.

Synthetic data. All employee and survey data were synthetically generated for demonstration purposes. The results therefore do not represent actual employee experiences, organizational conditions, or observed turnover behavior. The relationships identified in the analysis should be interpreted as properties of the simulated dataset rather than estimates of real-world employee behavior.

Cross-sectional survey design. The analysis primarily examines employee-experience measures and intent to stay at the survey-response level. Although the project includes survey-wave information, the analyses presented here do not establish whether changes in employee-experience over time lead to changes in intent to stay.

Associational, not causal, findings. The regression results identify statistical associations between employee-experience constructs and intent to stay. They do not demonstrate that improving a particular construct would cause employees to be more likely to remain with the organization.

Ordinal outcome. Intent to stay is measured on a five-point scale. Linear regression was used as a practical modeling approximation for this portfolio analysis, but the outcome is ordinal rather than truly continuous.

Model scope. The predictive model includes six employee-experience constructs and evaluates whether adding job level, manager status, work arrangement, and tenure provides additional explanatory value. Other factors that could influence intent to stay—such as compensation, external

labor-market conditions, workload changes, career opportunities, or individual circumstances—are not included.

Generalizability. Because the organization, survey responses, and data-generating process are simulated, the findings cannot be generalized to other organizations or employee populations. In a real employee-listening program, conclusions would need to be evaluated against the organization's actual workforce, survey methodology, response patterns, and operational context.

Conclusion

This analysis demonstrates an end-to-end employee-listening and organizational-health analytics workflow, from survey measurement and data-quality assessment through organizational analysis and predictive modeling.

Across 4,344 usable survey responses, employee-experience scores were generally centered near the midpoint of the five-point scale, with relatively small differences at the organization and business-unit levels. Department-level analysis revealed greater variation in selected areas, particularly Growth & Development and Manager Effectiveness.

Intent to Stay was positively associated with all six employee-experience constructs. Engagement showed the strongest association, followed by Manager Effectiveness and Belonging. The final predictive model explained approximately 58% of the variance in Intent to Stay on the held-out test data, while adding job level, manager status, work arrangement, and tenure did not significantly improve model fit.

Taken together, the analysis illustrates how employee-listening data can be translated into evidence-based organizational insights while incorporating measurement validation, predictive evaluation, and privacy-conscious reporting. The results also demonstrate the importance of moving beyond organization-wide averages to identify areas where more targeted follow-up may be warranted. Because the dataset is synthetic, the primary value of this project is demonstrating the analytical approach and decision-making framework rather than making claims about actual employee behavior. In a real organizational setting, the same workflow could support leaders in identifying employee-experience priorities, directing further investigation, and evaluating whether subsequent listening efforts show meaningful change.